

3.3.6

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Question:

Construct a triangle ABC with side $\mathbf{BC} = 6\text{cm}$, $\angle B = 45^\circ$, $\angle A = 105^\circ$

Solution:

Let position vector of $\mathbf{B}$ be $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and a, b, c be the sides opposite to the vertices A, B and C

Given $a=6\text{cm}$;

$$\mathbf{C} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \quad (0.1)$$

$$\mathbf{A} = \begin{pmatrix} c \cos \angle B \\ c \sin \angle B \end{pmatrix} \quad (0.2)$$

$$\mathbf{A} = \begin{pmatrix} \frac{c}{\sqrt{2}} \\ \frac{c}{\sqrt{2}} \end{pmatrix} \quad (0.3)$$

$$\angle A + \angle B + \angle C = 180^\circ \quad (0.4)$$

$$\therefore \angle C = 30^\circ \quad (0.5)$$

in $\triangle ABC$

$$b \cos \angle C + c \cos \angle B = a \quad (0.6)$$

$$b \sin \angle C - c \sin \angle B = 0 \quad (0.7)$$

$$\begin{pmatrix} \cos \angle C & \cos \angle B \\ \sin \angle C & -\sin \angle B \end{pmatrix} \begin{pmatrix} b \\ c \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \quad (0.8)$$

using augmented matrix

$$\left(\begin{array}{cc|c} \cos \angle C & \cos \angle B & 6 \\ \sin \angle C & -\sin \angle B & 0 \end{array} \right) \quad (0.9)$$

$$\left(\begin{array}{cc|c} \sqrt{3}/2 & 1/\sqrt{2} & 6 \\ 1/2 & -1/\sqrt{2} & 0 \end{array} \right) \quad (0.10)$$

$$R_2 \longrightarrow \sqrt{3}R_2 - R_1$$

$$\left(\begin{array}{cc|c} \sqrt{3}/2 & 1/\sqrt{2} & 6 \\ 0 & -\sqrt{3} - 1/\sqrt{2} & -6 \end{array} \right) \quad (0.11)$$

$$\implies c = \frac{6\sqrt{2}}{\sqrt{3} + 1} \quad (0.12)$$

$$\therefore \mathbf{A} = \begin{pmatrix} 6(\sqrt{3} - 1) \\ 6(\sqrt{3} - 1) \end{pmatrix} \quad (0.13)$$

